"Climate change isn’t just a looming crisis—it’s a burning threat to our planet now!"

EGYPT'S CLIMATE

AND THE ROLE OF

SUSTAINABILITY

TEAM : DIVE THEN
THRIVE

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01 Introduction

Egypt has a unique and diverse climate — from rich soil types to varying temperatures and high solar radiation. This natural diversity opens doors to incredible opportunities in agriculture, renewable energy, and sustainable development

However, unlocking this potential starts with one key factor: understanding our climate accurately and in-depth. That's where our project comes in — transforming climate data into actionable insights.

Purpose of Study

- The idea started with our awareness of the serious impacts of climate change on Egyptian society and economy.

- As data analysts, we aim to use data to highlight the climate change crisis, its severity, its rapid progression, and critical insights, empowering environmental specialists to develop solutions or mitigate its impact.

Problem Statement

Despite the richness of Egypt's environment, the lack of accurate, up-to-date, and connected climate data leads to poor decision-making, particularly in agriculture. Many Egyptian farmers still rely on outdated information or personal intuition. They often lack tools to:

- Track and understand weather changes
- Analyze how climate affects soil quality
- Choose the best crops for each season This gap in accessible climate intelligence also limits renewable energy planning and sustainable building design.

Objectives

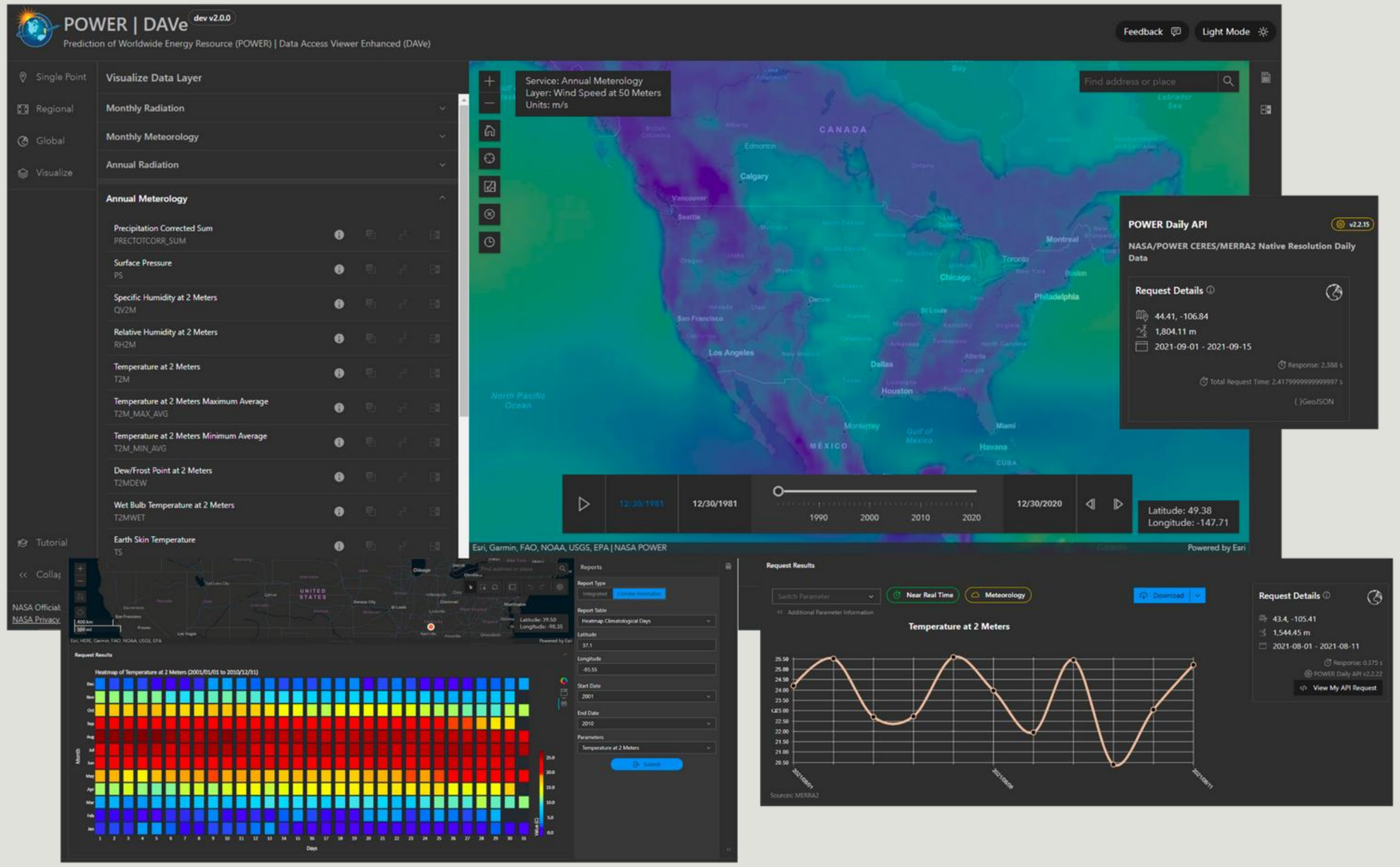
Our project aims to:

- Decode Egypt’s climate using real data and advanced analytics
- Empower farmers with smart agricultural insights
- Help identify optimal regions for solar and wind energy
- Support sustainable urban planning decisions
- Align with Egypt Vision 2030 in food security, clean energy, and sustainable land use


DATA
DATA
COLLECTION

02 Data Collection

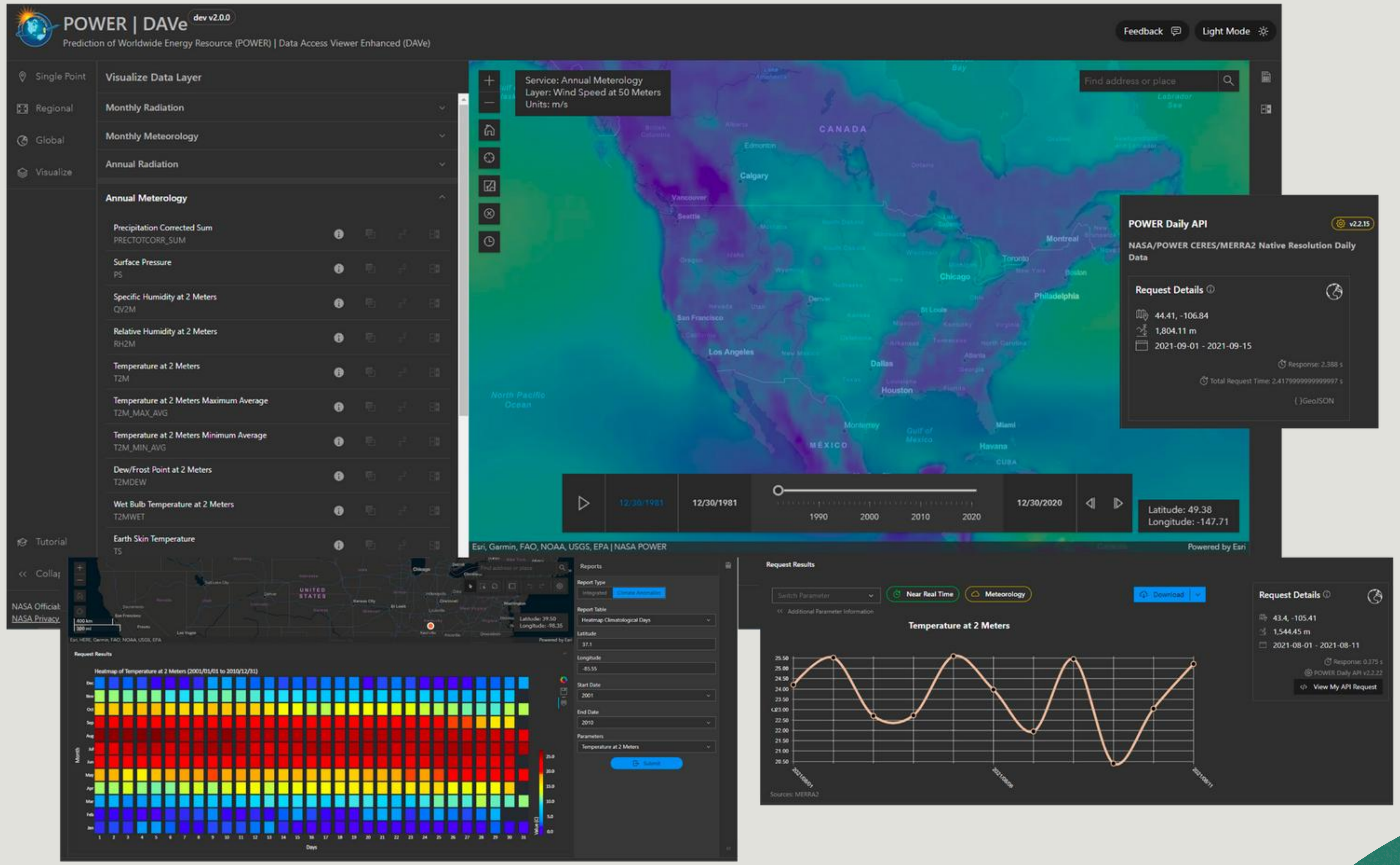
- **Source:** NASA Power Dataset
- **48 feature** where chosen from the data based on availability and importance over last 5 years.
- **Download Method:** Python requests from API
- Data was downloaded in files containing one year period with one feature per file.
- The data has to be downloaded into two rectangles covering Egypt.



48 Feature x 5 years x 2 Regions = Total 480 file

02 Data Collection

- 2 features where discarded for having no data.
- The files of each feature was appended together.
- Then, the feature files where merged into 2 major files.
- We couldn't merge those two files normally because of the changes in location ranges.
- Each of the two files where massive with **hundreds of thousands of rows** each.



03 Data Preprocessing

Combining Data

We plotted the points of the large file and the small file on the map and found that the largest difference between any two adjacent points was small. So, we performed proximity and used point-to-point approximation for the small dataset.

And finally we combined them into a single file with unified points and the attributes from both files.

Data Exploration

1- Using MySQL

We explored data with MySQL, but the large dataset caused slow queries, system overload, memory issues, and scalability limits made it an impractical solution for our large dataset.

Data Exploration

2- Using the **Ydata-profiling library** in Python for **Exploratory data analysis (EDA)** provides:

- A quick summary of our dataset

    includes the number of rows, columns, and data types.

- Identifies issues

    like such as missing values, duplicates, and outliers.

- The calculation of statistics

    like mean, median, standard deviation, minimum, maximum

Problems & Solutions in data

1- Outliers

We found outliers in the data, but they have meaning, due to unusual climate conditions so we cannot remove them

.

Solution:

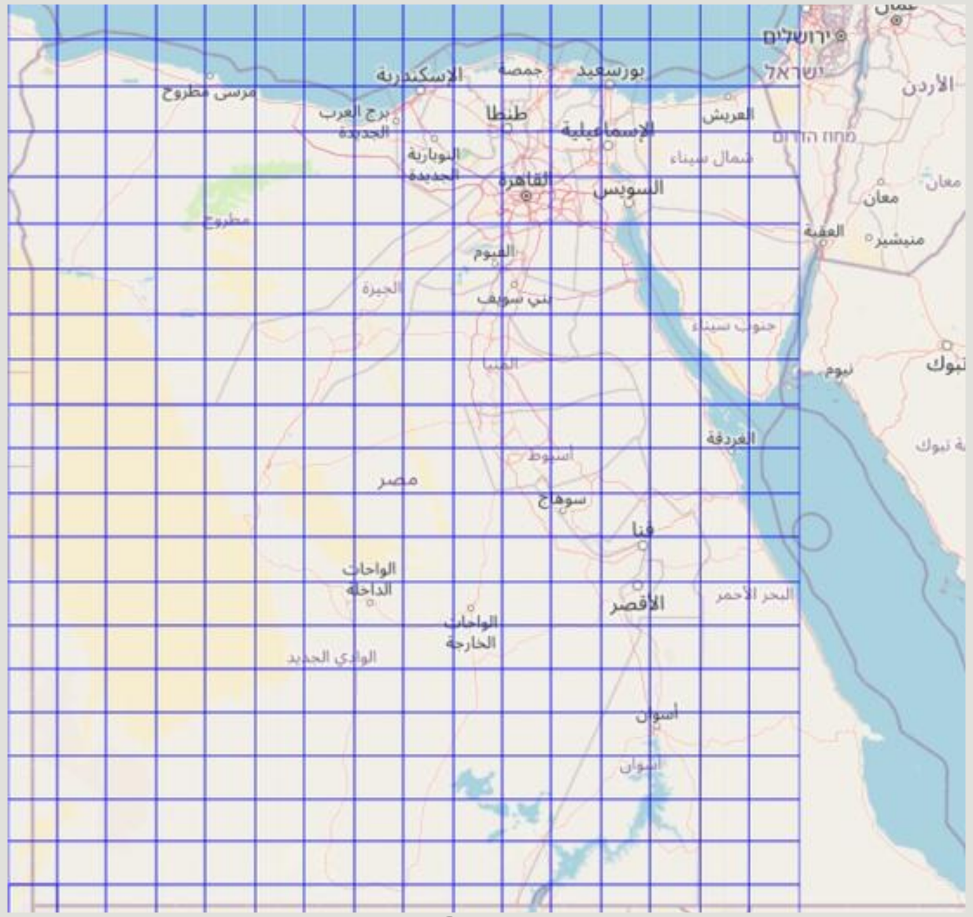
We handled outliers using normalization, and sometimes we used percentiles 5 and 95 in some analyses where calculating outliers wasn’t suitable.

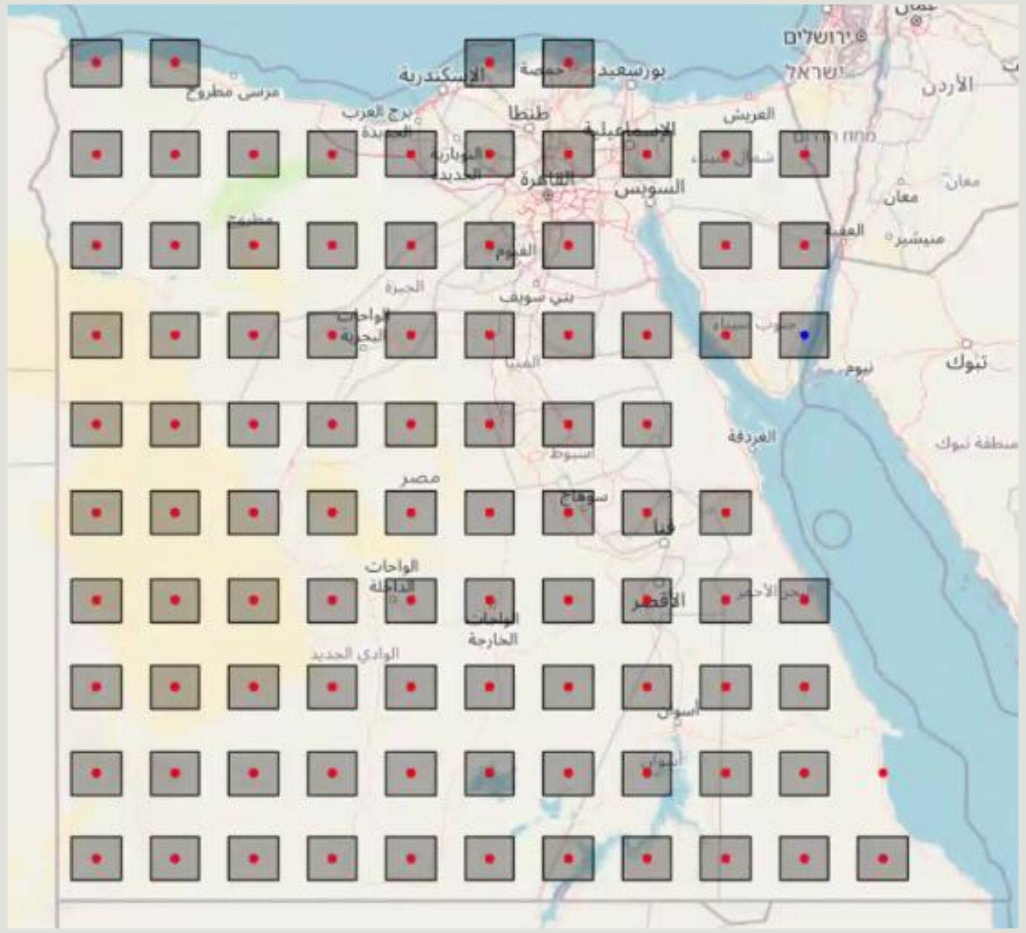
In other cases, we kept the outliers based on their significance and our understanding of the data’s needs for specific purposes.

Problems & Solutions in data

2- Missing Values

We found missing values and we could detect the reason that the Egypt boundaries used in NASA's data were inaccurate, including points outside Egypt and in the seas.





We utilized shapefiles with Folium maps to identify and manually remove points outside Egypt’s borders, particularly those in the sea, as they caused numerous attributes to have missing values in our dataset.

Geospatial and Temporal Data Enhancement

1-Filtering Original Datasets

Filtered the datasets to keep only points within Egypt’s borders.

2- Assigning City and Governorate

Used a shapefile to assign each point its nearest city and

. governorate, adding "City" and "GOV" columns.

3- Adding Temporal Features (Extraction feature)

Created a "Date" column from YEAR and DOY, then extracted "Week

. Number" and "Month Number" as new columns for temporal analysis.

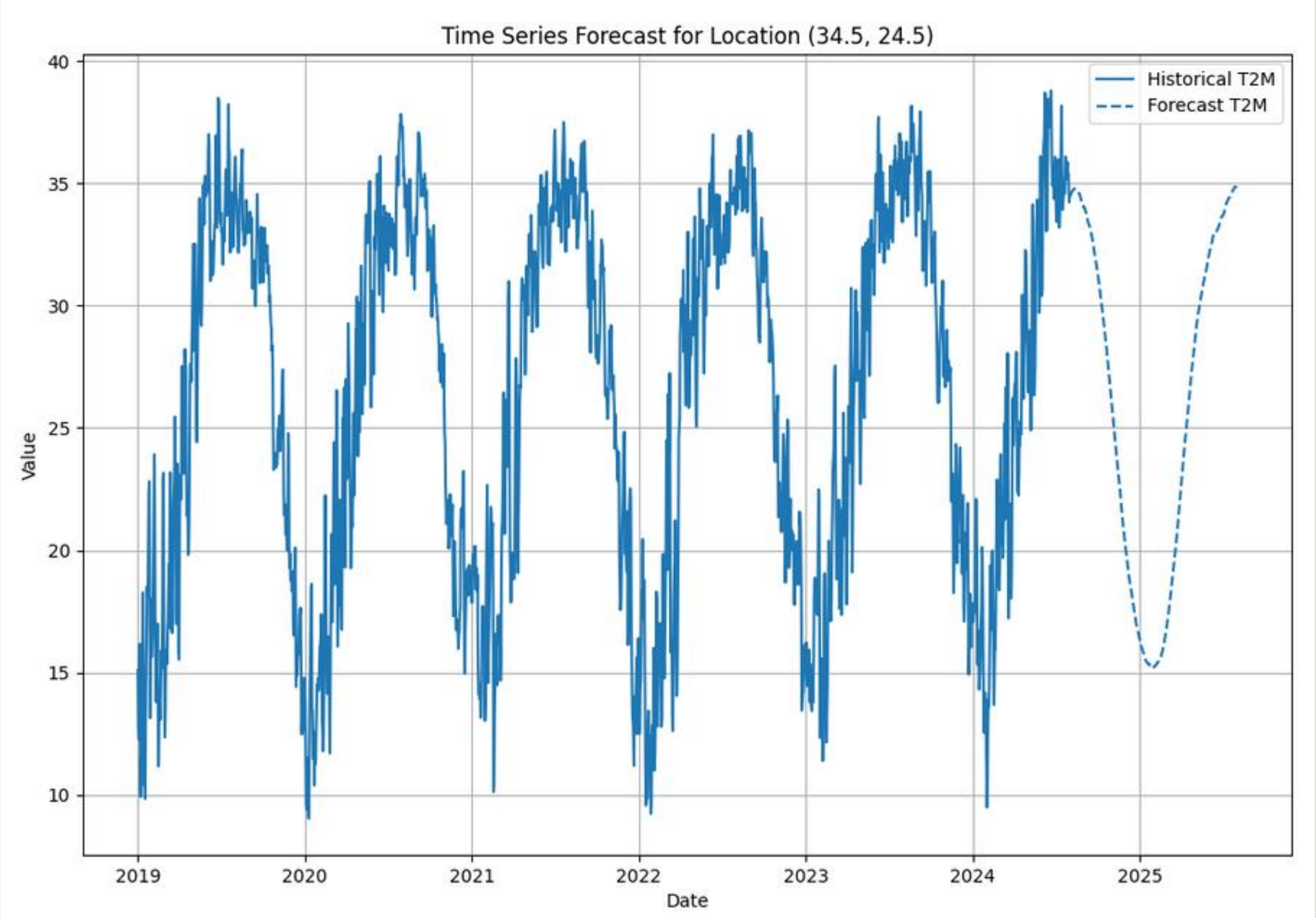
04 Data Analysis

Streamlit Website

- Data Summaries where collected into one website for quick access.
- **Tool:** Python Streamlit Library
- The website had three tabs:
  - Metadata and Visualizations
  - Weather Predictions
  - Agricluster Analysis

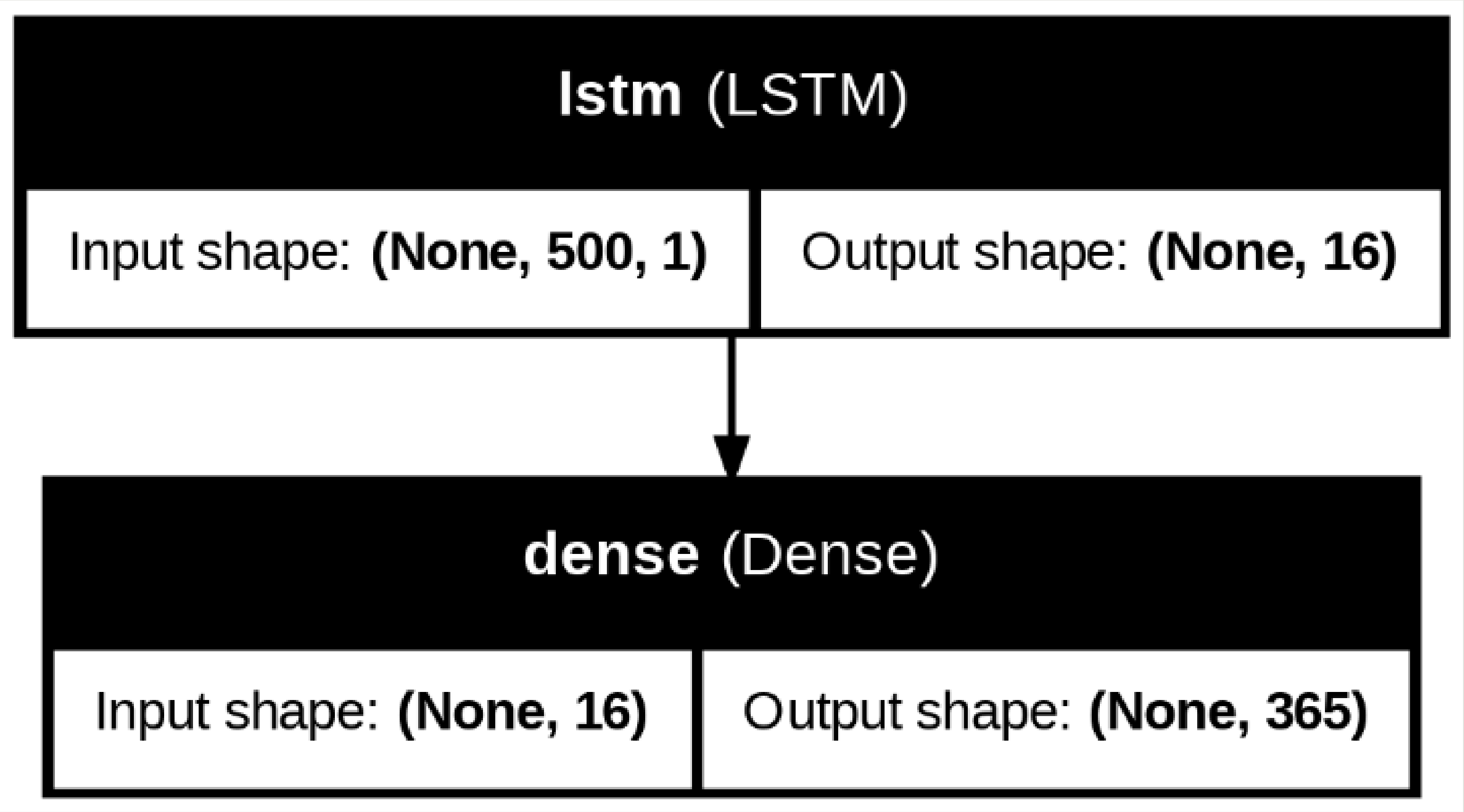
05 Weather Prediction

- **Objective:** Forecast 13 climate features for 365 days across 90 Egypt locations.
- **Purpose:** The predicted data have been used for further applications and analysis.
- **Dataset:** Sub-dataset contains 13 chosen features to focus on.
- **Approach:** Train LSTM models, export, and predict using Kaggle.



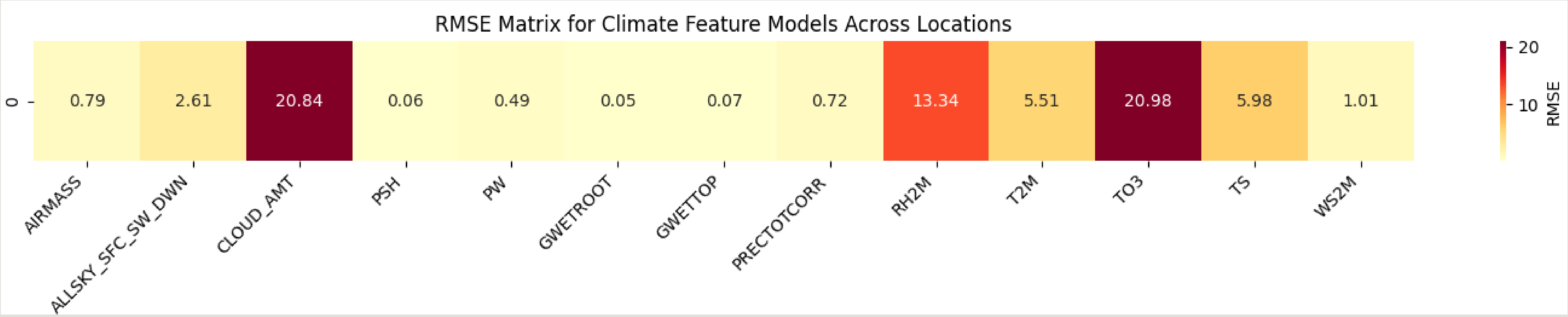
LSTM Model

- **Input:** 500 timesteps, 1 feature.
- **LSTM:** 16 units to capture patterns.
- **Output:** 365-day (whole Year) forecast.
- Trained with MSE loss, saved on Kaggle.



Performance and Results

- Evaluated with RMSE on test predictions.
- RMSE heatmap shows performance across 13 features and 90 locations.



- Generated 365-day forecasts, exported to forecasted_data.csv.

CSV export enables recent real-world applications.

06 Data Clustering

1- The Goal

-To present climate data clearly and accessibly for environmental researchers and data scientists exploring localized climate patterns

-support agricultural consultants in identifying region-specific climate conditions

-enabling them to study and utilize the data easily for informed plant selection and farming decisions.

2- How It Was Implemented and Achieved

-We used K-Means Clustering to divide Egypt into 4 clusters based on Silhouette Score.

-aggregated data per cluster per season for accuracy.

-displayed key location data (e.g., distance to Nile, elevation).

-showed percentiles (5 and 95) to exclude outliers.

-listed suitable plants per cluster and season.

-provided a table with plant attributes and simple solutions.

3. Sources and Tools Used

-We sourced elevation data from Open Elevation API

-Distances to coasts from Natural Earth

-Nile River data from OpenStreetMap

-land cover from FAO

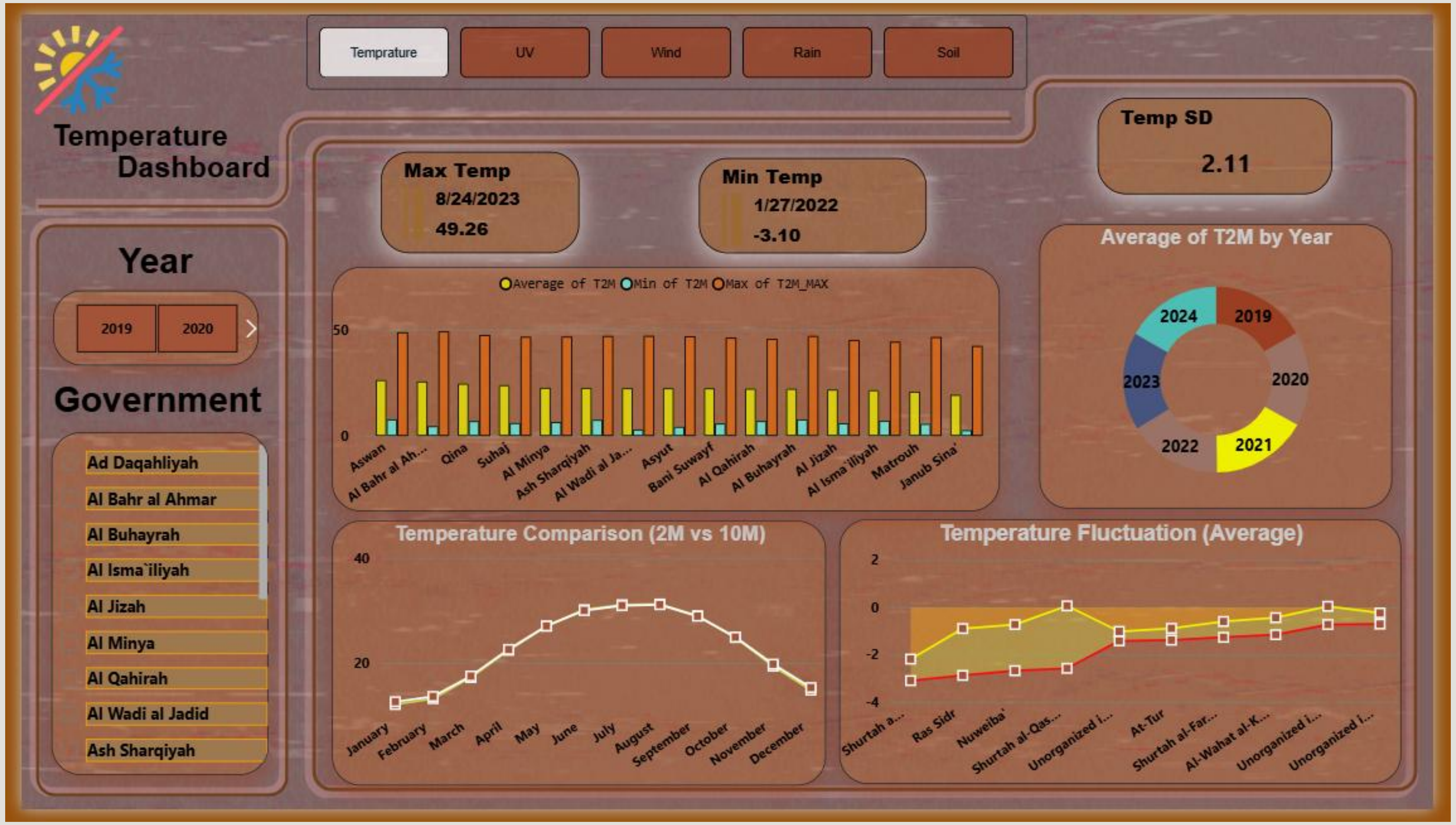
-We used K-Means Clustering and Silhouette Score cluster division.

-Plant data from FAO.

-We employed HTML, JavaScript, Python, and CSS to present results clearly and user-friendly.

07 Data Visuallization

- **Used Tools:** Microsoft Power Bi
- **Dashboards are :**
    a. Temperature
    b. Sun Radiation
    c. Winds
    d. Rain
    e. Soil



Alignment with Egypt Vision 2030

Egypt Vision 2030

Egypt Vision 2030 focuses on improving the quality of life for Egyptian citizens and enhancing their standard of living in all aspects of life.

Egypt Vision 2030 also prioritizes addressing the impacts of climate change through an integrated and sustainable ecosystem that enhances resilience and the ability to confront natural hazards.

Alignment with Egypt Vision 2030

ALIGNMENT WITH EGYPT VISION 2030

✅ Environmental Pillar

Supports the sustainable use of Egypt's natural resources in agriculture and energy.

✅ Economic Pillar

Boosts agricultural productivity, reduces waste, and promotes renewable energy investment.

✅ Urban Development Pillar

Enables smart, climate-adaptive building and city planning through accurate weather-based insights.

THANK YOU!